

CLINICAL TRIAL TREATMENT EFFECTIVENESS AND SAFETY ANALYSIS

Pharmaceutical Analytics | Clinical Trial Intelligence | Power BI Reporting

Prepared as a portfolio-grade analytics case study

Tools Used: Python, PostgreSQL, Power BI

1. Executive Context

Pharmaceutical organizations rely on clinical trial analytics to determine whether a therapy is delivering meaningful patient benefit while maintaining an acceptable safety profile. Trial stakeholders must continuously evaluate treatment response, survival outcomes, adverse events, dosage patterns, patient demographics, tumor characteristics, and high-risk patient groups.

This project simulates a real-world pharma analytics use case where patient-level chemotherapy data is converted into actionable insights for trial monitoring, treatment comparison, and safety review. The final output is designed to support business intelligence users, clinical operations teams, and healthcare analytics stakeholders.

2. Business Problem

The key business problem is to identify which chemotherapy regimens and patient segments are associated with stronger treatment response, better survival outcomes, and higher adverse event risk, so that clinical stakeholders can monitor trial performance more effectively.

Without structured analysis, decision makers may struggle to compare treatment groups, detect safety patterns, understand patient-risk distribution, or communicate clinical trial performance through clear KPIs. This can reduce the effectiveness of trial monitoring and delay insight-driven decision-making.

3. Project Objective

The objective of this project is to build an end-to-end clinical trial analytics workflow that cleans patient-level data, performs exploratory and SQL-based analysis, and presents the most important clinical KPIs through a one-page Power BI dashboard.

4. Key Analytical Questions

1. What is the overall treatment response rate across the trial population?
2. Which chemotherapy regimen has the strongest response-rate performance?
3. How are patients distributed across tumor stages, cancer types, and treatment regimens?
4. Which age groups show higher severe side-effect risk?
5. How do survival outcomes vary across tumor stages?
6. What proportion of patients experience neutropenia or severe nausea?
7. How can clinical KPIs be presented in a dashboard for quick stakeholder review?

5. Dataset Scope

Area	Details
Dataset Size	52,321 patient records and 17 source columns
Patient Profile	Age, sex, BMI, smoking status
Disease Profile	Cancer type, genetic mutation, tumor stage, tumor size, metastasis status
Treatment Profile	Chemotherapy regimen, dosage in mg/m2, cycles completed
Safety Profile	Nausea severity, neutropenia, severe side-effect flag
Outcome Profile	Tumor response and overall survival months

6. Success Metrics

Effectiveness Metrics

Overall response rate, complete response rate, partial response rate, average survival months, response by regimen.

Safety Metrics

Severe side-effect rate, neutropenia rate, nausea severity by age group, adverse event pattern by regimen.

Patient Segmentation

Age group, sex, cancer type, tumor stage, metastasis status, genetic mutation, smoking status.

Dashboard Performance

One-page KPI view with slicers, comparison visuals, and stakeholder-friendly storytelling.

7. Tools and Technology Stack

Tool	Business Purpose
Python	Data cleaning, transformation, feature engineering, EDA, and KPI calculation
PostgreSQL / pgAdmin 4	Database loading, SQL querying, and metric validation
Power BI	Dashboard design, slicers, KPI cards, and executive visual reporting
Jupyter Notebook	Reusable analytical documentation and code workflow

8. Stakeholders

- Clinical operations managers monitoring trial progress
- Pharmaceutical business analysts evaluating treatment performance
- Drug safety teams reviewing adverse event patterns
- Medical affairs teams communicating clinical insights
- Business intelligence teams building reporting solutions for healthcare users

9. Expected Business Value

The project provides a structured way to evaluate treatment effectiveness and safety using patient-level trial data. It enables quick comparison of chemotherapy regimens, highlights patient groups requiring closer monitoring, and translates complex clinical information into dashboard-driven business insights.

10. Final Deliverables

- Analysis-Ready Clinical Trial Dataset
- Python Analytical Workflow
- PostgreSQL Query Layer
- Executive Power BI Dashboard
- Business Analysis Documentation